

C02 AT3 LEVEL 2

The image shows a C++ programming environment with two different code snippets and their outputs.

Top Screenshot:

Your Code

```
#include<iostream>
#include<cstring>
using namespace std;
int isAnagram(char str1[],char str2[])
{
    int count[256]={0};
    if(strlen(str1)!=strlen(str2))
        return 0;
    for(int i=0;str1[i]!='\0';i++){
        count[(unsigned char)str1[i]]++;
        count[(unsigned char)str2[i]]--;
    }
    for(int i=0;i<256;i++){
        if(count[i]!=0)
            return 0;
    }
    return 1;
}
int main()
{
    char str1[100],str2[100];
    cin>>str1;
    cin>>str2;
    if(isAnagram(str1,str2))
        cout<<str1<<" and "<<str2<<" are Anagrams ";
    else
        cout<<str1<<" and "<<str2<<" are not Anagrams ";
    return 0;
}
```

Program Input (stdin)

```
listen
silent
```

Output

```
listen and silent are Anagrams
```

Bottom Screenshot:

Occurrences of 'ana': 2

Test Case 1: Enter the main string: Banana Enter the substring to count: ana

Test Case 2: Enter the main string: abcdabcabdcabc Enter the substring to count: abc

Test Case 3: Enter the main string: Hello World World Hello Hello Enter the substring to count: World

Your Code

```
#include <iostream>
using namespace std;
int main()
{
    string str="Banana";
    string sub="ana";
    int count=0;
    for(int i=0;i<str.length()- 2;i++){
        if(str[i]=='a' && str[i+1]=='n' && str[i+2]=='a')
            count++;
    }
    cout<<" occurrences of 'ana'=" <<count;
    return 0;
}
```

Program Input (stdin)

```
Banana
ana
```

Output

```
occurrences of 'ana'= 2
```

Sample Output:
Sorted array: 9 5 4 2 1

Your Code

```
#include<iostream>
using namespace std;
int main()
{
    int n,a[100],temp;
    cin>>n;
    for(int i=0;i<n;i++){
        cin>>a[i];
    }
    for(int i=0;i<n;i++){
        for(int j=i+1;j<n;j++){
            if(a[i]<a[j]){
                temp=a[i];
                a[i]=a[j];
                a[j]=temp;
            }
        }
    }
    for(int i=0;i<n;i++)
        cout<<a[i]<<" ";
    return 0;
}
```

Program Input (stdin)

```
5
4 2 9 1 5
```

Output

```
9 5 4 2 1
```

Run

Save

Next →

Your Code

```
#include<iostream>
#include<cstring>
using namespace std;
int isAnagram(char str1[],char str2[])
{
    int count[256]={0};
    if(strlen(str1)!=strlen(str2))
        return 0;
    for(int i=0;str1[i]!='\0';i++){
        count[(unsigned char)str1[i]]++;
        count[(unsigned char)str2[i]]--;
    }
    for(int i=0;i<256;i++){
        if(count[i]!=0)
            return 0;
    }
    return 1;
}
int main()
{
    char str1[100],str2[100];
    cin>>str1;
    cin>>str2;
    if(isAnagram(str1,str2))
        cout<<str1<<" and "<<str2<<" are Anagrams ";
    else
        cout<<str1<<" and "<<str2<<" are not Anagrams ";
    return 0;
}
```

Program Input (stdin)

```
listen
silent
```

Output

```
listen and silent are Anagrams
```

Run

Save

